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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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CR MILES, P.C.
CRAIG R. MILES
405 MASON COURT, SUITE 119
FORT COLLINS, CO 80524

EXAMINER
THANH, QUANG D

ART UNIT	PAPER NUMBER
3771	

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12/23/2014	PAPER

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Office Action Summary	Application No. 14/316,668	Applicant(s) GEORGE ET AL.	
	Examiner Quang D. Thanh	Art Unit 3771	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☐ Responsive to communication(s) filed on _____.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL**. 2b)☒ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5)☒ Claim(s) *1-3,5-7,9-12,14,16-20,22,25,29,30,33-36,38,40,41,43 and 44* is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6)☐ Claim(s) _____ is/are allowed.

7)☒ Claim(s) *1-3,5-7,9-12,14,16,18-20,22,25,29,30,33-36,38,40 and 43* is/are rejected.

8)☒ Claim(s) *17,41 and 44* is/are objected to.

9)☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on *6/26/14* is/are: a)☒ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date *6/26/14,11/15/14,12/8/14*.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date. _____ .

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. This office action is responsive to the amendments filed on 7/24/14 and 8/11/14 . As directed by the amendment: no claim has been amended; claims 4,8, 13, 15, 21, 23-24, 26-28, 31-32, 37, 39, 42 and 45-142 have been canceled. Thus, claims 1-3, 5-7, 9-12, 14, 16-20, 22, 25, 29-30, 33-36, 38, 40-41 and 43-44 are presently pending in this application.

Claim Rejections - 35 USC § 102

3. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

4. Claims 1-3, 5-7 and 10 are rejected under 35 U.S.C. 102 (a) (1) as being anticipated by Franz et al. (US 7,766,858).

5. Regarding claim 1, Franz discloses an external ear canal pressure regulation device (abstract), comprising: providing a first fluid flow generator 10 (fig. 1) capable of generating a first fluid flow (“air pushing device”, col. 3, lines 36-40); a first earpiece 48 (fig. 12) having a first axial earpiece conduit 71 (fig. 12) which communicates between a first earpiece first end and a first earpiece second end (col. 4, lines 57-58), said first axial earpiece conduit fluidicly coupled to said first fluid flow generator (fig. 12), said first

earpiece 48 having a first compliant (to provide “a good generally airtight seal”, see col 4, lines 55-60) earpiece external surface 70 configured to sealably engage a first external ear canal of a first ear as a first barrier between a first external ear canal pressure and an ambient pressure (“airtight seal”, col 4, lines 55-60).

6. Regarding claim 2, Franz discloses said first fluid flow generator 10 generates said first fluid flow between said first fluid flow generator and said first axial earpiece conduit, said first fluid flow having a first fluid volume in a range of between 0 milliliters to about 20 milliliters (col 5, lines 32-47, Franz teaches 7 milliliters as the volume of the ear cavity and increasing pressure by 215 daPa thus the volume of fluid flow would necessarily be less than 7 milliliters).

7. Regarding claim 3, Franz discloses said first fluid volume comprises a first pre-selected fluid volume in said range of between 0 milliliters to about 20 milliliters (col 5, lines 32-47, Franz teaches 7 milliliters as the volume of the ear cavity and increasing pressure by215 daPa thus the volume of fluid flow would necessarily be less than 7 milliliters).

8. Regarding claim 5, Franz discloses said first fluid flow generator capable of generating a first pressure differential between said first external ear canal pressure and said ambient pressure (col 5, lines 32-47).

9. Regarding claim 6, Franz discloses said first pressure differential having a first pressure differential amplitude in a range of between 0 kilopascals to about 50 kilopascals (col 5, lines 32-47, 215 decapascals=2.15 kilopascals).

10. Regarding claim 7, Franz discloses said first pressure differential amplitude comprises a first pre-selected pressure differential amplitude in said range of between 0 kilopascals to about 50 kilopascals (col 5, ln 32-47, 2.15 decapascals=2.15 kiloPascals).

11. Regarding claim 10, Franz discloses said first fluid flow generator capable of generating a first pressure differential amplitude oscillation which reciprocally drives said first fluid flow between a first fluid flow first direction and a first fluid flow second direction in said first axial earpiece conduit (col 2, lines 25-28 and col 5, lines 32-47, the positive and negative pressures taught by Franz would inherently produce fluid flows in opposite directions).

Claim Rejections - 35 USC § 103

12. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

13. Claims 9, 11-12, 14, 16 are rejected under 35 U.S.C. 103 as being unpatentable over Franz in view of Engvall et al. (US 6,629,938 B1).

14. Regarding claim 9, Franz discloses all the claimed features except for a first pressure differential amplitude selection element; and a first fluid flow generator controller responsive to operation of said first pressure differential amplitude selection element to regulate operation of said first fluid flow generator to achieve said first pre-

selected pressure differential amplitude. However, Engvall also teaches a device for pressure regulation within an ear canal (abstract) and teaches a first pressure differential amplitude selection element (col 5, lines 51-67); and a first fluid flow generator controller 16 responsive to operation of said first pressure differential amplitude selection element to regulate operation of a first fluid flow generator to achieve said first pre- selected pressure differential amplitude (col 5, lines 54-58). Therefore, it would have been obvious before the effective date of the claimed invention to one of ordinary skill in the art to modify the device in the Franz's reference, to include a first pressure differential amplitude selection element; and a first fluid flow generator controller responsive to operation of said first pressure differential amplitude selection element to regulate operation of said first fluid flow generator, as suggested and taught by Engvall, for the purpose of achieving said first pre- selected pressure differential amplitude to automate the pressure regulation systems as taught by Engvall (col. 5, lines 51-58).

15. Regarding claim 11, Engvall also teaches a device for pressure regulation within an ear canal (abstract) and teaches differential amplitude oscillation having a first pressure differential amplitude oscillation frequency in a range of between 0 Hertz to about 10 Hertz (col 3, ln 38-55).

16. Regarding claim 12, Engvall also teaches a first pressure differential amplitude oscillation frequency comprises a first pre-selected pressure differential amplitude oscillation frequency in said range of between 0 Hertz to about 10 Hertz (col 3, ln 38-55 of Engvall).

17. Regarding claim 14, Engvall teaches a first pressure differential amplitude selection element (col 5, ln 51-67); and a first fluid flow generator controller 16 responsive to operation of said first pressure differential amplitude selection element to regulate operation of a first fluid flow generator to achieve said first pre-selected pressure differential amplitude (col 5, lines 54-58).

18. Regarding claim 16, Franz discloses all the claimed features except for a first pressure sensor which generates a first pressure sensor signal which varies based upon change in said first pressure differential amplitude; and a first pressure sensor signal analyzer including a first pressure differential amplitude comparator which functions to compare said first pre-selected pressure differential amplitude to said first pressure differential amplitude, said first pressure sensor signal analyzer generating a first pressure differential amplitude compensation signal, said first fluid flow generator controller responsive to said first pressure differential amplitude compensation signal to control said first fluid flow generator to achieve said first pre- selected pressure differential amplitude. However, Engvall also teaches a device for pressure regulation within an ear canal (abstract) and teaches a first pressure sensor 8 which generates a first pressure sensor signal which varies based upon change in said first pressure differential amplitude; and a first pressure sensor signal analyzer 16 including a first pressure differential amplitude comparator which functions to compare said first pre-selected pressure differential amplitude to said first pressure differential amplitude (a comparison with the acceptable interval is implied by the col 5, lines 54-60), said first pressure sensor signal analyzer generating a first pressure differential amplitude

compensation signal 13 (col 5, lines 56-58), said first fluid flow generator controller 16 responsive to said first pressure differential amplitude compensation signal to control said first fluid flow generator to achieve said first pre-selected pressure differential amplitude (col 5, lines 54-60). Therefore, it would have been obvious before the effective date of the claimed invention to one of ordinary skill in the art to modify the device in the Franz's reference, to include first pressure sensor and first pressure sensor analyzer as taught by Engvall in the ear canal pressure regulator of Franz, as suggested and taught by Engvall, for the purpose of providing and ensuring the patient's safety during use so that the pressure level does not exceed a physiological acceptable interval (col 5, ln 48- 60 of Engvall)

19. Claims 18-20 are rejected under 35 U.S.C. 103 as being unpatentable over Franz in view Densert et al. (US 6,159,17).

20. Regarding claim 18, Franz discloses all the claimed features except for a fluid flow temperature regulator fluidicly coupled between said first fluid flow generator and said first axial earpiece conduit which operates to regulate a first fluid flow temperature of said first fluid flow. However, Densert also teaches a device for pressure regulation within an ear canal (abstract) and teaches a fluid flow temperature regulator fluidicly coupled between a fluid flow generator and an axial earpiece conduit which operates to regulate a first fluid flow temperature of the fluid flow (col 6, lines 18-39). Therefore, it would have been obvious before the effective date of the claimed invention to one of ordinary skill in the art to modify the device in the Franz's reference, to include a fluid flow temperature regulator in the ear canal pressure regulator, as suggested and taught

by Densert, for the purpose of avoiding vertigo in a patient (col 6, lines 18-39 of Densert).

21. Regarding claim 19, Densert also teaches the first fluid flow temperature in a range of between about 10 degrees Celsius to about 50 degrees Celsius (37 degrees Celsius, col 6, lines 20).

22. Regarding claim 20, Densert also teaches a first preselected fluid flow temperature in said range of between about 10 degrees Celsius to about 50 degrees Celsius (37 degrees Celsius, col 6, line 20).

23. Claims 22, 25, 29 and 30 are rejected under 35 U.S.C. 103 as being unpatentable over Franz in view Oyadiran et al. (US 2011/0224493).

24. Regarding claim 22, Franz discloses all the claimed features except for a second earpiece having a second axial earpiece conduit which communicates between a second earpiece first end and a second earpiece second end, said second axial earpiece conduit fluidicly coupled to said first fluid flow generator, said second earpiece having a second compliant earpiece external surface configured to sealably engage a second external ear canal of a second ear as a second barrier between a second external ear canal pressure and said ambient pressure. However, Oyadiran is also related to earpieces for medical treatment and teaches duplicating the components of a first earpiece for a second earpiece (abstract). Further, it would have been obvious to a person of ordinary skill in the art wherein the second earpiece would have the same structure as the first earpiece as disclosed by Franz, per the opinion, as set forth above in claim 1. Therefore, it would have been obvious before the effective date of the

claimed invention to one of ordinary skill in the art to modify the device in the Franz‘s reference, to duplicate the components of the Franz's first earpiece of Franz to provide a second earpiece with identical components as suggested by Oyadiran, for the purpose of providing simultaneous treatment for both ears as suggested by the abstract of Oyadiran.

25. Regarding claim 25, Franz discloses all the claimed features except for a second fluid flow generator capable of generating a second fluid flow; and a second earpiece having a second axial earpiece conduit which communicates between a second earpiece first end and a second earpiece second end, said second axial earpiece conduit fluidicly coupled to said second fluid flow generator, said second earpiece having a second compliant earpiece external surface configured to sealably engage a second external ear canal of a second ear as a second barrier between a second external ear canal pressure and said ambient pressure. However, Oyadiran is also related to earpieces for medical treatment and teaches duplicating the components of a first earpiece for a second earpiece (abstract). Further, it would have been obvious to a person of ordinary skill in the art wherein the second earpiece would have the same structure as the first earpiece as disclosed by Franz, per the opinion, as set forth above in claim 1. Therefore, it would have been obvious before the effective date of the claimed invention to one of ordinary skill in the art to modify the device in the Franz‘s reference, to duplicate the components of the Franz's first earpiece of Franz to provide a second earpiece with identical components as suggested by Oyadiran, for the

purpose of providing simultaneous treatment for both ears as suggested by the abstract of Oyadiran.

26. Regarding claim 29, Franz in view Oyadiran discloses the subject matter of claims 25, as described above, in which the second fluid flow generator is capable of generating a second pressure differential between said second external ear canal pressure and said ambient pressure (col 5, lines 32-47 of Franz).

27. Regarding claim 30, Franz in view of Oyadiran discloses the subject matter of claim 29, as described above, in which said second pressure differential having a second pressure differential amplitude in a range of between 0 kilopascals to about 50 kilopascals (col 5, lines 32-47 of Franz, 215 decapascals=2.15 kilopascals).

28. Claims 33-36, 38 and 40 are rejected under 35 U.S.C. 103 as being unpatentable over Franz/Oyadiran et al. and further in view of Engvall.

29. Regarding claim 33, Franz/Oyadiran discloses all the claimed features except for a second pressure differential amplitude selection element; and a second fluid flow generator controller responsive to operation of said first pressure differential amplitude selection element to regulate operation of said first fluid flow generator to achieve said first pre-selected pressure differential amplitude. However, Engvall also teaches a device for pressure regulation within an ear canal (abstract) and teaches a pressure differential amplitude selection element (col 5, lines 51-53); and a fluid flow generator controller 16 responsive to operation of said pressure differential amplitude selection element to regulate operation of a fluid flow generator to achieve a pre-selected pressure differential amplitude (col 5, lines 54-58). Therefore, it would have been

obvious before the effective date of the claimed invention to one of ordinary skill in the art to modify the device in the Franz‘s reference, to include a second pressure differential amplitude selection element; and a second fluid flow generator controller responsive to operation of said second pressure differential amplitude selection element to regulate operation of said second fluid flow generator, as taught by Engvall to achieve said second pre-selected pressure differential amplitude of Franz/Oyadiran in order to automate the pressure regulation systems as taught by Engvall (col. 5, lines 51-58).

30. Regarding claim 34, Franz/Oyadiran and in view of Engvall discloses the subject matter of claim 33, as described above, wherein said second fluid flow generator capable of generating a second pressure differential amplitude oscillation which reciprocally drives said second fluid flow between a second fluid flow first direction and a second fluid flow second direction in said second axial earpiece conduit (col 2, lines 25-28, col 5, ln32-47, the positive and negative pressures taught by Franz would inherently produce fluid flows in opposite directions).

31. Regarding claims 35-36, Engvall also teaches a pre-selected pressure differential amplitude oscillation frequency in a range of between 0 Hertz to about 10 Hertz (col 3, lines 38-55 of Engvall). Therefore, it would have been obvious before the effective date of the claimed invention to one of ordinary skill in the art to modify the device in the Franz‘s reference, to provide said second pre-selected pressure differential amplitude oscillation in a range of between 0 Hertz to about 10 Hertz as taught by Engvall, for the purpose of providing a suitable oscillation frequency selected as a treatment for

Meniere's disease (col 1, lines 61-63 of Engvall) which is also a same concern of Franz (abstract of Franz).

32. Regarding claim 38, Franz/Oyadiran in view of Engvall discloses the subject matter of claim 36, as described above, including a second pressure differential amplitude oscillation frequency selection element (col 5, lines 51-53 of Engvall), said second fluid flow generator controller 16 responsive to operation of said second pressure differential amplitude oscillation frequency selection element to regulate operation of said second fluid flow generator to achieve said second preselected pressure differential amplitude oscillation frequency (col 5, lines 54-58 of Engvall).

33. Regarding claim 40, Franz/Oyadiran discloses all the claimed features except for a second pressure sensor which generates a second pressure sensor signal which varies based upon change in said second pressure differential amplitude; and a second pressure sensor signal analyzer including a second pressure differential amplitude comparator which functions to compare said second pre-selected pressure differential amplitude to said second pressure differential amplitude, said second pressure sensor signal analyzer generating a second pressure differential amplitude compensation signal, said second fluid flow generator controller responsive to said first pressure differential amplitude compensation signal to control said second fluid flow generator to achieve said first pre-selected pressure differential amplitude. Engvall also teaches a device for pressure regulation within an ear canal (abstract) and teaches a pressure sensor 8 which generates a pressure sensor signal which varies based upon change in said first pressure differential amplitude; and a pressure sensor signal analyzer 16

including a pressure differential amplitude comparator which functions to compare said first pre-selected pressure differential amplitude to said first pressure differential amplitude (a comparison with the acceptable interval is implied by the col 5, lines 54-60), said pressure sensor signal analyzer generating a first pressure differential amplitude compensation signal 13 (col 5, lines 56-58), said fluid flow generator controller 16 responsive to said first pressure differential amplitude compensation signal to control said first fluid flow generator to achieve said first pre- selected pressure differential amplitude (col 5, lines 54-60). Therefore, it would have been obvious before the effective date of the claimed invention to one of ordinary skill in the art to modify the device in the Franz's reference, to include second pressure sensor and second pressure sensor analyzer as taught by Engvall in the second ear canal pressure regulator of Franz/Olyadiran, for the purpose of providing and ensuring the patient's safety during use so that the pressure level does not exceed a physiological acceptable interval (col 5, lines 48- 60 of Engvall)

34. Claim 43 are rejected under 35 U.S.C. 103 as being unpatentable over Franz/Oyadiran et al. and further in view of Densert.

35. Regarding claim 43, Franz/Oyadiran discloses all the claimed features except for a fluid flow temperature regulator fluidicly coupled to said first fluid flow and said second fluid flow which operates to regulate a first fluid flow temperature of said first fluid flow and a second fluid flow temperature of said. second fluid flow. However, Densert also teaches a device for pressure regulation within an ear canal (abstract) and teaches a fluid flow temperature regulator fluidicly coupled between a fluid flow generator and an

axial earpiece conduit which operates to regulate a fluid flow temperature of the fluid flow (col 6, lines 18-39). Therefore, it would have been obvious before the effective date of the claimed invention to one of ordinary skill in the art to modify the device in the Franz's reference, to include a fluid flow temperature regulator in the ear canal pressure regulator, as suggested and taught by Densert, for the purpose of avoiding vertigo in a patient (col 6, lines 18-39 of Densert).

Double Patenting

36. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the claims at issue are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the reference application or patent either is shown to

be commonly owned with this application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO internet Web site contains terminal disclaimer forms which may be used. Please visit http://www.uspto.gov/forms/. The filing date of the application will determine what form should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to http://www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

37. Claims 1-3,5-7,9-12,14,16,18-20,22,25,29,30,33-36,38,40 and 43 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1, 3, 4, 6, 9,12, 22, 25, 26 and 28 of copending Application No. 14/292,469 in view of Franz, Engvall, Oyadiran and/or Densert. Although the claims at issue are not identical, they are not patentably distinct from each other because the claims of said copending application, in view of any one or combination of the references listed above, teach essentially all the claimed features.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Allowable Subject Matter

38. Claims 17, 41 and 44 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Quang D. Thanh whose telephone number is (571)272-4982. The examiner can normally be reached on Monday-Friday 9am-5pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Justine Yu can be reached on (571) 272-4835. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Quang D. Thanh/
Primary Examiner, Art Unit 3771